Transcendental Syntax

The dynamics of logic programs and tilings, applied to Linear Logic

Team LoVe – LIPN Université Sorbone Paris Nord

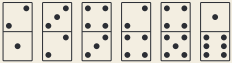
Boris ENG

Stellar Resolution

From tiles to logic programs

Wang tiles

Hao Wang (1961)

Dominos. 

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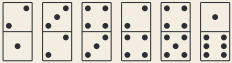
Dominos. 

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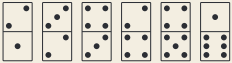
Dominos. 

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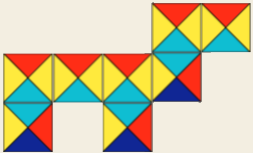
Tiling. 

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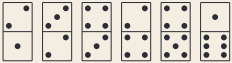
Dominos. 

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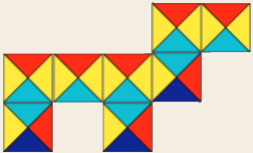
Tiling. 

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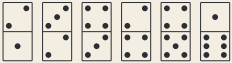
Dominos. 

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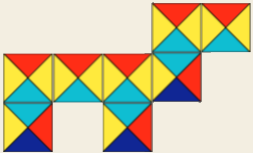
Tiling. 

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Generalisations. different matchability (e.g DNA computing), higher dimensions (e.g $\mathbf{Z}^3$).

Flexible tiles

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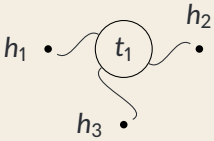

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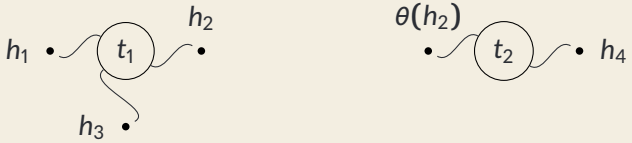

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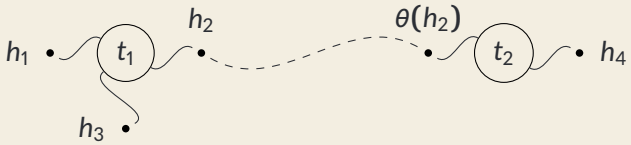

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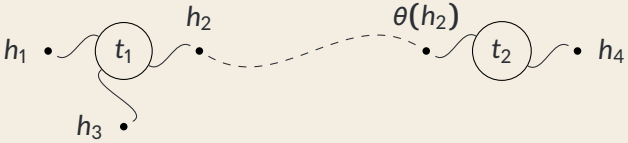

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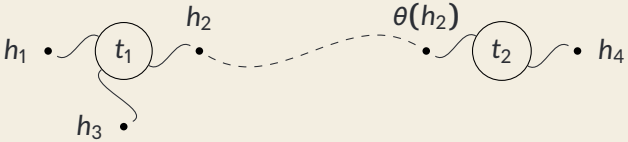

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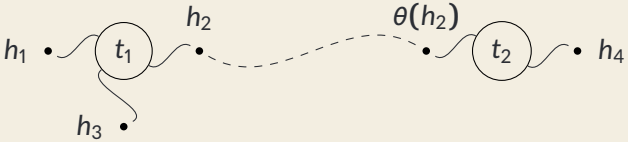

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- Related to `NTIME` classes.
  "*Complexity classes for self-assembling flexible tiles*" (Jonoska et al.).

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Flexible tiles on first-order terms.

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↳ for $x \doteq f(x) \simeq_\alpha y \doteq f(x)$ we have $\theta = y \mapsto f(x)$

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Stars and constellations

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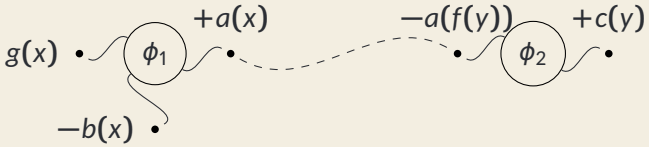

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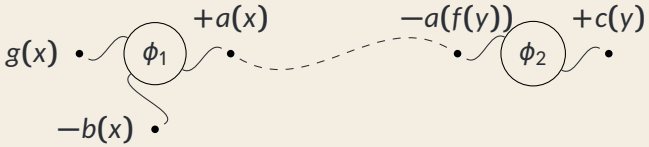

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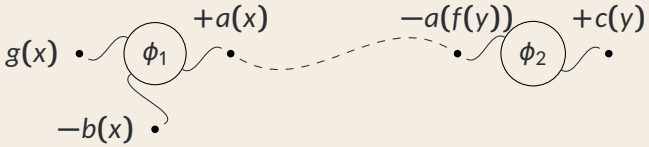

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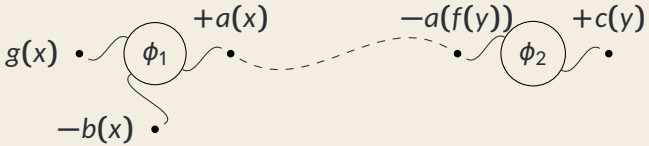

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Tiles	Resolution	Stellar Resolution
	Atom $A = A(t), \neg A(t)$	Ray $r = +a(t), -a(t)$
Tile	Clause $C = A_1 \vee \ldots \vee A_n$	Star $\phi = [r_1, \ldots, r_n]$
Tile set	Program $P = C_1 \wedge \ldots \wedge C_m$	Constellation $\Phi = \phi_1 + \ldots + \phi_m$
Tiling	Inference tree	Diagram

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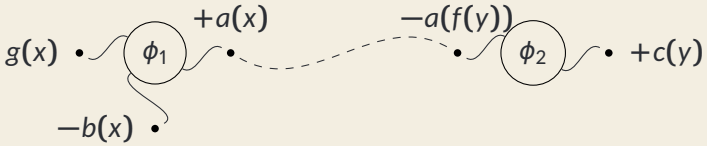

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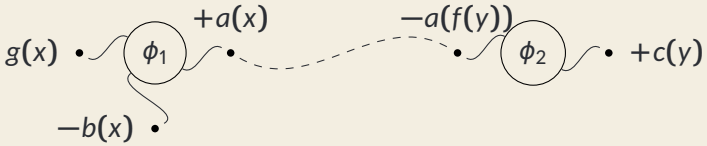

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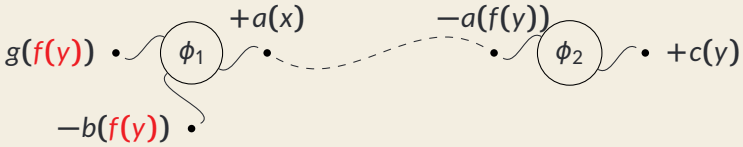

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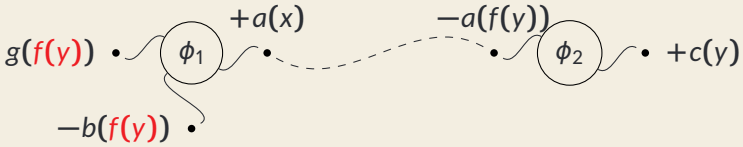

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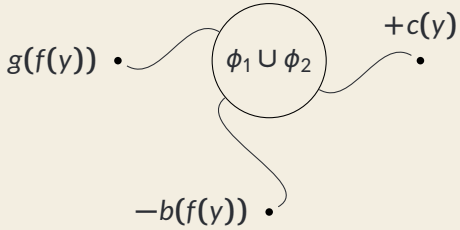

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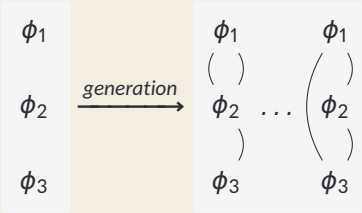

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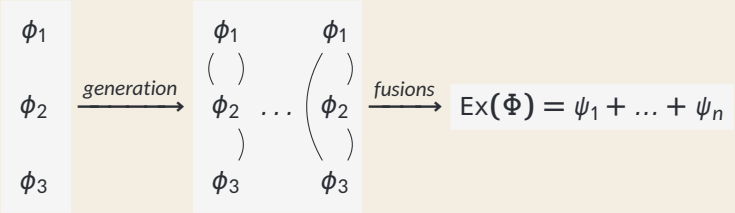

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- We want the diagrams to be saturated (impossible to extend).
- We also want them to be correct (no unification error).

Stellar Resolution

Few examples

Unary addition by **logic programming** :

$$[+add(0, y, y)] + [-add(x, y, z), +add(s(x), y, s(z))] + [-add(s^n(0), s^m(0), r), r]$$

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All other diagrams fail, hence $\mathrm{Ex}(\boldsymbol{\Phi}) = [s^4(0)]$.

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$$[+val(0)] + [+val(1)] + [+not(1,0)] + [+not(0,1)] + [+or(0,0,0)] + [+or(0,1,1)] + [+or(1,0,1)] + [+or(1,1,1)]$$

Stellar Resolution

Few examples

Boolean circuits as hypergraph+dynamics : $X \vee \neg X$

$$\frac{-val(x), X(x)}{+c_1(x)}\ ;$$

$$\frac{-c_1(x)}{+c_2(x), +c_3(x)}\ ;$$

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$$\mathrm{Ex}(\Phi) = [X(0), R(1)] + [X(1), R(1)]$$

Extensible to arithmetic circuits

Transcendental Syntax

Proofs as tilings

Transcendental Syntax

Motivations

Explain (linear) logic from its computational behaviour.

Transcendental Syntax

Motivations

Explain (linear) logic from its computational behaviour. By finite means !

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Answers/Analytic adequate computational space : stellar resolution.

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"*This can only be a reconstruction, which means that we roughly know what we are aiming at.*" (Transcendental Syntax I).

Transcendental Syntax

MLL proof-structures

An alternative representation of proofs as hypergraphs :

Transcendental Syntax

MLL proof-structures

An alternative representation of proofs as hypergraphs :

Formulas/Vertices. $A, B := X \mid A \otimes B \mid A \parr B$

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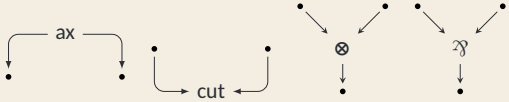

Rules/Hyperedges. Axiom Cut Tensor Par

Transcendental Syntax

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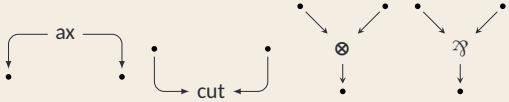

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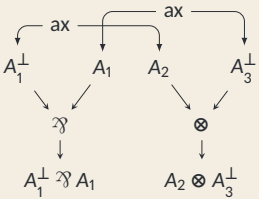

Transcendental Syntax

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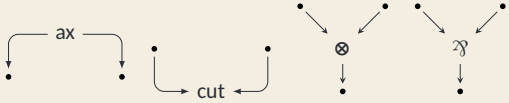

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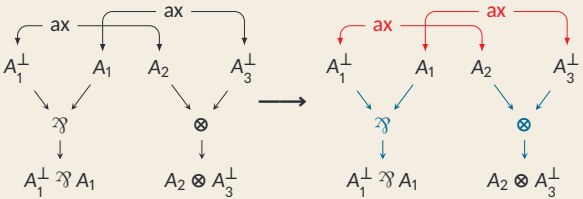

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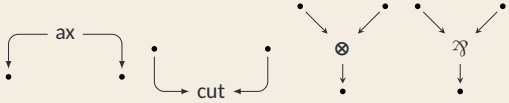

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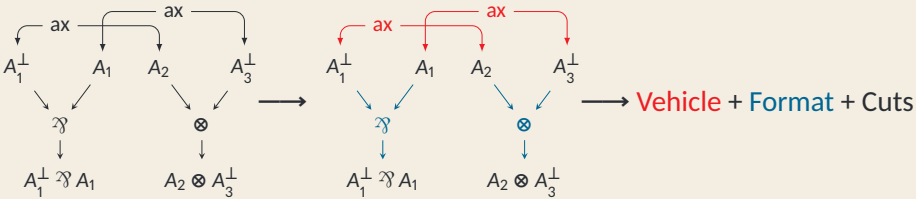
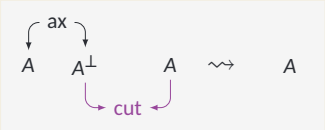


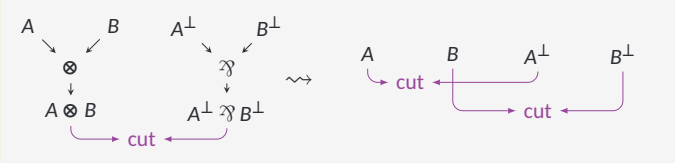
Vehicle + Format + Cuts

Transcendental Syntax

The computational content of proof-structures

Cut-elimination procedure :





Transcendental Syntax

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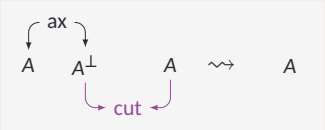

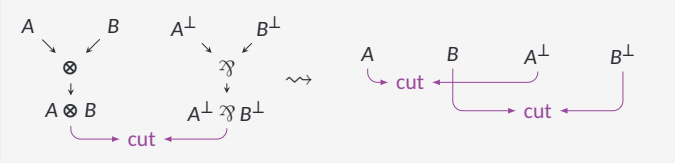

Geometry of Interaction :

Transcendental Syntax

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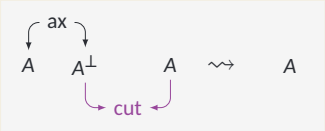
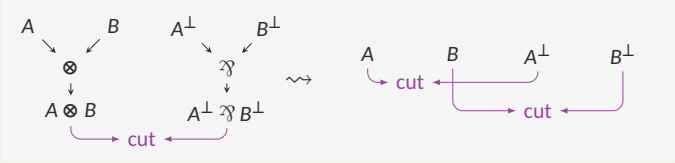


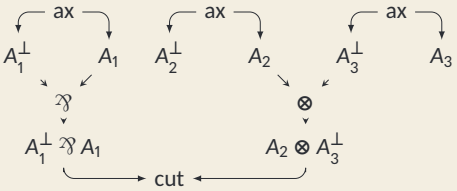


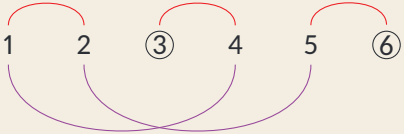
Geometry of Interaction :

Basically a computation of maximal paths in a graph.

Transcendental Syntax

Simulation of cut-elimination





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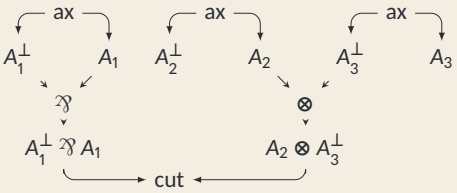

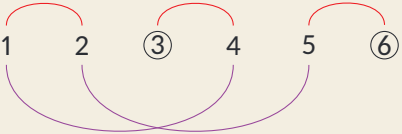

$+c.p_{A_1^\perp \parr A_1}(\mathtt{l}x);\ \ +c.p_{A_1^\perp \parr A_1}(\mathtt{r}x);$

$+c.p_{A_2^\perp}(x);\ \ +c.p_{A_2 \otimes A_3^\perp}(\mathtt{l}x);$

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Transcendental Syntax

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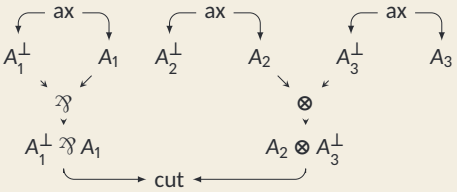

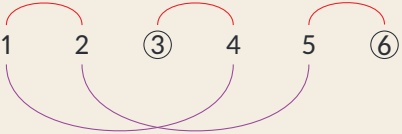

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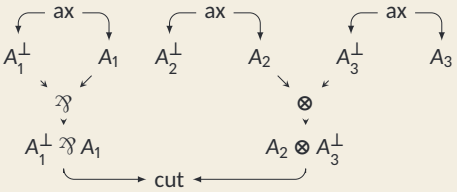

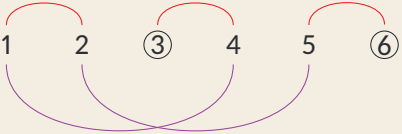

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$+c.p_{A_2^\perp}(x);\ +c.p_{A_2 \otimes A_3^\perp}(\mathtt{lx})\,;$

$+c.p_{A_2 \otimes A_3^\perp}(\mathtt{rx});\ +c.p_{A_3}(x)\,;$

$-c.p_{A_1^\perp \parr A_1}(\mathtt{lx});\ -c.p_{A_2 \otimes A_3^\perp}(\mathtt{lx})\,;$

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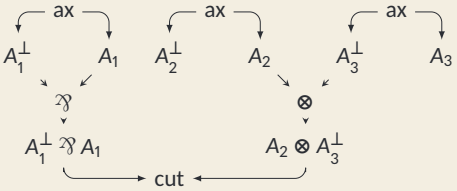

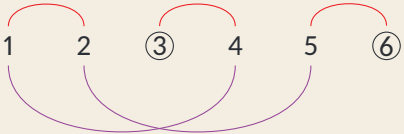

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Transcendental Syntax

The logical content of proof-structures

Only some proof-structures are "logically correct".

Transcendental Syntax

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Danos-Regnier **correctness criterion** : testing the vehicle against several Tests $\in$ Format.

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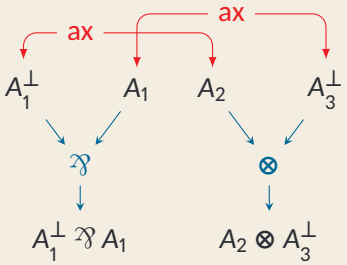

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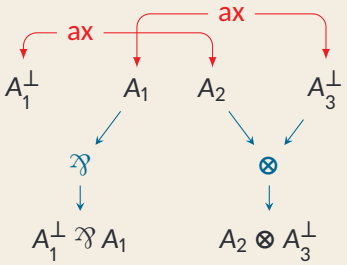

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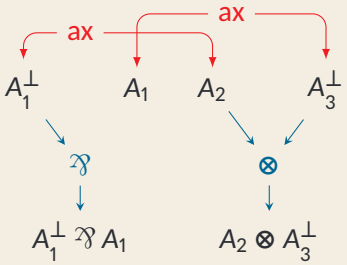

Transcendental Syntax

Simulation of correctness

Stars $\equiv$ Oriented hyperedges

$+t.p_{A \otimes B}(\mathtt{l}x);\ \ +t.p_{A^\perp \parr B^\perp}(\mathtt{l}x)\,;$

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$$+t.p_{A\otimes B}(\mathtt{lx});\quad +t.p_{A^\perp \parr B^\perp}(\mathtt{lx})\,;$$

$$[\,\frac{-t.p_{A\otimes B}(\mathtt{lx})}{+c.q_A(x)}\,]\,;$$

$$[\,\frac{-t.p_{A\otimes B}(\mathtt{rx})}{+c.q_{A^\perp}(x)}\,]\,;$$

$$[\,\frac{-c.q_A(x)\quad -c.q_B(x)}{+c.q_{A\otimes B}(x)}\,]\,;$$

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$$\frac{-c.q_{A^\perp}(x)}{}\,;$$

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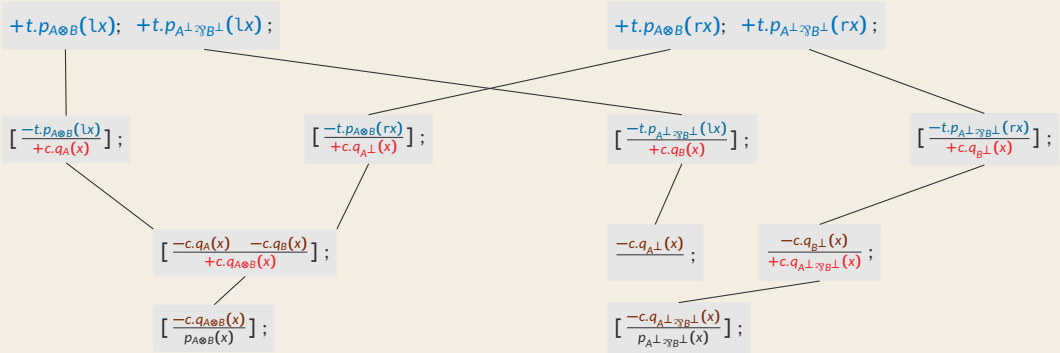

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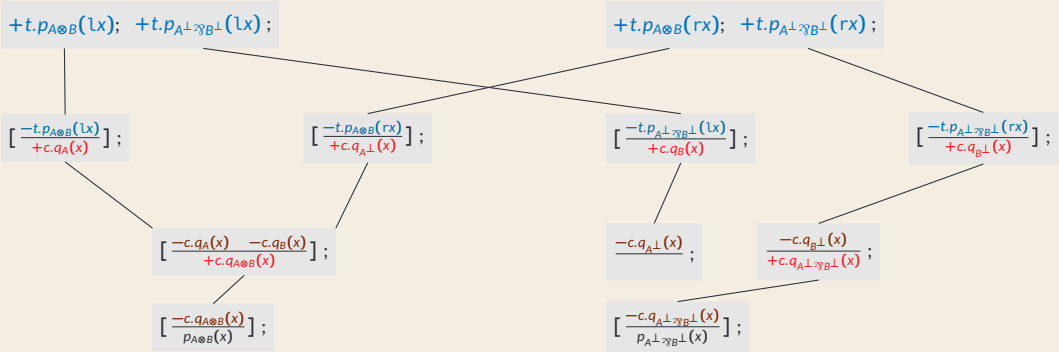

Property of the tiling : **logically correct** iff for all test, the normal form is a single star.

Transcendental Syntax

Typing and formulas

Use of techniques from "linear" realisability.

Transcendental Syntax

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Other constructions $\mathbf{A} \parr \mathbf{B} := (\mathbf{A}^\perp \otimes \mathbf{B}^\perp)^\perp$, $\qquad \mathbf{A} \multimap \mathbf{B} := \mathbf{A}^\perp \parr \mathbf{B}$.

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Pre-types description of a behaviour $\mathbf{A} = \{\Phi_i\}_{i \in I}$.

Orthogonality Choose a definition of "good interaction" $\Phi \perp \Phi'$.

Dual pre-type $\mathbf{A}^\perp$ set of "good partners" $\{\Phi \mid \forall \Phi_A \in \mathbf{A}, \Phi \perp \Phi_A\}$.

Types $\mathbf{A} = \mathbf{A}^{\perp\perp}$ closed interaction.

Tensor $\mathbf{A} \otimes \mathbf{B} := \{\Phi_A \uplus \Phi_B \mid \Phi_A \in \mathbf{A}, \Phi_B \in \mathbf{B}\}^{\perp\perp}$.

Other constructions $\mathbf{A} \parr \mathbf{B} := (\mathbf{A}^\perp \otimes \mathbf{B}^\perp)^\perp$, $\qquad \mathbf{A} \multimap \mathbf{B} := \mathbf{A}^\perp \parr \mathbf{B}$.

$\Phi_1 \perp \Phi_2 \iff |\mathrm{Ex}(\Phi_1 \uplus \Phi_2)| = 1$: captures MLL formulas.

Transcendental Syntax

Conclusion

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  - ↳ equality as linear equivalence (not predicate !)
- Logic programs and functional programs, unified ?

Future and related works

(actually a call for help)

Hypergraphings

Hypergraphs with dynamics

Computation with hypergraphs :

Model	**Vertices**	**Hyperedges**
Boolean circuits	addresses	gates
Proof-nets	addresses	rules
Constellations	terms	stars
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↳ categorical framework ? Operads, string diagrams, hypergraph categories, frobenius algebras, ...

Stellar Resolution and Automata Theory

Dependency graph $\mathfrak{D}(\Phi)$: relations of matchability within a constellation Φ.

$-add(0, y, y);$ —— $+add(x, y, z);$ $-add(s(x), y, s(z));$ —— $+add(s^n(0), s^m(0), r);$ $r;$

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Diagram (formally) : graph homomorphism $\delta : G \longrightarrow \mathfrak{D}(\Phi)$.

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Run in finite automata : path $\longmapsto$ state graph

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↳ automaton for stellar execution ?